US-China Decoupling and Board Composition: Evidence from Chinese Listed Firms

Kenneth Khoo[*] kenneth.khoo@nus.edu.sg James Si Zeng[†] jszeng@hku.hk

Abstract

When the United States imposed sanctions and investment bans on Chinese firms, U.S.-connected directors of Chinese listed companies became personally exposed to legal and compliance risk under the U.S. Persons Rule. The timing and targeting of this shock were set by U.S. national-security policy and were largely external to firm governance, so we treat it as quasi-exogenous and ask how Chinese boards adjusted. Because director nationality is not reported in standard corporate data, we construct it with a classification pipeline that anchors an auditable three-prong U.S.-nexus definition in structured CSMAR records and supplements it with web-scraped biographies labeled by a large-language model, validated across the mainland A-share and Hong Kong markets. We find that exposed mainland firms shed roughly two percentage points of U.S.-nexus director share, an effect concentrated among independent directors and realized through non-renewal at term end. The exit appears even at firms under no firm-specific prohibition, which we interpret as compliance chill operating through individual liability. Substitution is market-specific: mainland boards recruit domestically credentialed professionals while Hong Kong boards recruit non-U.S. internationals, and the response is selective at exposed firms but broader at those explicitly sanctioned.

[*]Assistant Professor, National University of Singapore, Faculty of Law.
[†]Associate Professor, The University of Hong Kong, Faculty of Law.

1 Introduction

In recent years, prominent Chinese listed firms have watched directors with American passports, American work histories, and American professional licenses leave their boards as Washington and Beijing have pulled their economies apart. Over this period the United States and China have undergone a marked process of economic and financial decoupling, and a growing literature documents its consequences for trade, innovation, and firm value (Egger and Zhu, 2020; He et al., 2022). A fundamental question nonetheless remains unanswered: through what internal governance mechanisms do geopolitical shocks reshape firms? This Article addresses that question by examining how U.S.–China decoupling has reshaped board composition in Chinese listed firms, and by asking who replaces the directors who depart. We document board reconstitution as a previously underexamined margin along which geopolitical fragmentation reaches the firm, a margin that operates through the market for director human capital as a channel distinct from the trade and financial channels emphasized in prior work.

Our analysis proceeds in two steps. First, we ask whether U.S. sanctions reshape board composition, that is, whether they reduce the share of directors with U.S. backgrounds on the boards of affected Chinese firms, and, when a seat is vacated, whether it is filled by a domestic professional, by a state-affiliated figure, or left empty. Second, we ask whether the same shock reshaped boards in an independent cross-market setting, namely Hong Kong-listed Chinese firms, where the foreign investor base is directly measurable and its response can be observed.

Answering these questions requires a variable that standard corporate-governance data do not report, namely the nationality and U.S.-nexus of individual directors. A second contribution of this Article is therefore a measurement strategy. We anchor an auditable three-prong U.S.-nexus definition—U.S. nationality or permanent residency, substantial U.S. work experience, and U.S. professional licensure—in structured CSMAR records, and, for directors whose connections are not verifiable from structured fields, we supplement it with web-scraped biographical text classified by a large-language model. To guard against classification error, we validate the procedure against an independent nationality classifier and apply it consistently across the mainland and Hong Kong markets.

We find that U.S.–China decoupling reduces the proportion of directors with U.S. backgrounds on the boards of Chinese listed firms by approximately two percentage points, roughly one-third of the pre-shock treated-firm mean. We further find that these directors are replaced predominantly by domestically credentialed professionals, and that departure occurs through passive non-renewal at term end, with forced mid-term resignation playing little role. Because the design is best suited to recovering the exit and leaves the subsequent staffing decision to descriptive analysis, we interpret the exit as causal—subject to the threats to identification we address in Section 10—while reading the replacement pattern as a descriptive characterization of board recomposition.

These results speak to several distinct questions. First, the design separates the U.S.-specific channel from a concurrent market-wide retreat of foreign directors from Chinese boards. A placebo decomposition shows that at exposed firms only U.S.-connected directors leave, while non-U.S.

foreign directors are unaffected on the mainland and actively recruited in Hong Kong; firms with non-U.S. foreign exposure but no U.S. connection, by contrast, lose those directors regardless of sanctions. To our knowledge, this is the first paper to isolate the sanctions-specific component of board de-internationalization from the broader Sino-Western governance fragmentation trend.

Second, this inquiry contributes to the growing literature on how geopolitical tensions affect corporate boards and firm governance (Liu et al., 2025; Tian et al., 2025). That literature is theoretically important because most corporate governance research treats board structure as the outcome of firm-level economic choices, such as the need for expertise, monitoring, or external resources (Adams et al., 2010). Geopolitical tensions, however, represent an external political shock that may alter board composition for reasons unrelated to firms' underlying economic fundamentals. By studying this channel, we deepen our understanding of how political frictions reshape internal governance arrangements through the allocation of top-level human capital.

Third, the exit appears across the full set of firms with a U.S.-connected director, including firms under no direct legal prohibition. We interpret this broad response as consistent with compliance chill, whereby heightened personal legal and reputational risk under the U.S. Persons Rule plausibly raised the cost of board service for U.S.-connected directors. Because the design identifies the exit itself while leaving the legal channel indirectly observed, we present compliance chill as the interpretation most consistent with the evidence and drawn from the observed pattern. Financial ownership, by contrast, appears to respond more narrowly: measurable reallocation surfaces only among explicitly sanctioned Hong Kong firms, where baseline international ownership was substantial, and it is absent both in the broader intent-to-treat samples and on the mainland, where pre-shock foreign exposure was negligible. While we read this contrast as suggestive of a higher threshold for capital flight than for director exit, we caution that the ownership nulls may partly reflect limited baseline exposure, which weakens any inference about investor non-response.

Fourth, the cross-market extension to Hong Kong provides an independent replication of the director-exit result in a distinct regulatory and ownership environment, and it shows that substitution is market-specific: mainland boards draw on a domestic talent pool, whereas Hong Kong boards draw on a deep non-U.S. international pool. Disentangling the U.S.-specific sanctions channel from the concurrent de-internationalization trend is, we believe, an important contribution to how board de-internationalization in Chinese firms is measured and interpreted.

These questions also carry policy implications. By documenting the exit of U.S.-connected directors and the domestic professionals who replaced them, this Article helps policymakers assess how geopolitical fragmentation reshapes corporate boards and how deep the domestic director-labor market is that absorbs such shocks. The pattern also points to an unintended consequence of extraterritorial sanctions. Because the directors who departed were disproportionately independent directors with international credentials—the type that a substantial literature associates with stronger monitoring and with constraining controlling-shareholder expropriation in Chinese firms (Zhu et al., 2016; Gong et al., 2021; Jiang et al., 2010)—a measure aimed at restricting U.S. capital flows may have incidentally removed a source of minority-shareholder protection from the affected boards, imposing the cost on domestic investors in a policy conflict between the sanctioning and sanctioned governments.

The remainder of the paper proceeds as follows. In Section 2, we describe the U.S. sanctions framework and the individual-liability channel; in Section 3, we review the related literature; and in Section 4, we set out the measurement strategy for U.S.-nexus board connections. Section 5 describes the data and empirical strategy, while Sections 6 and 7 report the exit of U.S.-nexus directors and examine board reconstitution, respectively. Section 8 presents the cross-market evidence and the compliance-chill threshold, Section 9 separates the sanctions channel from market-wide de-internationalization, and Section 10 discusses limitations.

2 Institutional Background: Sanctions and Individual Liability

The United States has enacted a series of executive orders that impose sanctions on Chinese listed firms. In 2019, President Trump signed Executive Order 13873, which authorized restrictions on transactions involving Chinese firms operating in the information and communications technology and services sector on national-security grounds. In 2020, the administration issued Executive Order 13959, which for the first time explicitly prohibited U.S. investors from investing in firms identified as "Communist Chinese military companies," that is, companies deemed to have ties to the Chinese military.

The Biden administration largely preserved and extended this policy direction. In 2021, it issued Executive Order 14032, which retained the earlier investment restrictions while broadening their scope. Building on these executive orders, the Office of Foreign Assets Control (OFAC) systematically established and updated the NS-CMIC list (i.e., the Non-SDN Chinese Military-Industrial Complex Companies List). According to official announcements, the first large-scale inclusion of Chinese firms under U.S. investment sanctions took place on November 12, 2020. OFAC subsequently undertook a major revision and expansion of the list on June 3, 2021, while the unified effective date for many of the investment prohibitions was set as August 2, 2021, from which point U.S. investors were formally prohibited from making new securities investments and capital allocations to the affected firms.

These prohibitions carry direct implications for directors who qualify as U.S. persons and serve on the boards of affected Chinese listed companies. The threshold concept is the definition of a "U.S. person," which the sanctions regime construes broadly to reach any U.S. citizen, lawful permanent resident, entity organized under U.S. law (including its foreign branches), and any individual physically present in the United States, irrespective of where the person resides or where the conduct occurs. A director who falls within this definition is bound by the prohibitions personally, independent of the nationality of the firm on whose board the director sits.

The operative prohibition, codified in the Chinese Military-Industrial Complex Sanctions Regulations (31 C.F.R. part 586), bars U.S. persons from the purchase or sale of any publicly traded securities of a listed NS-CMIC firm, together with derivatives of, and instruments designed to provide investment exposure to, such securities. Although board service as such is not enumerated among the prohibited acts, the individual-liability exposure that concerns us arises at the intersection of three features of the regime. First, the regulations separately prohibit any transaction that evades or avoids, has the purpose of evading or avoiding, or causes a violation of the prohibitions, together with conspiracies and attempts to do so, so that a U.S.-person director who approves or authorizes a covered securities transaction by the firm, such as an issuance, a buyback, or treasury dealing in covered instruments, may thereby cause or facilitate a violation even without trading on personal account. Second, equity-linked director compensation, including shares, options, or restricted units in a covered firm, can itself constitute the acquisition of a prohibited security. Third, because civil liability under the International Emergency Economic Powers Act attaches on a strict-liability basis, without proof of willfulness, a director's good-faith uncertainty about whether a particular board decision touches a covered security does not avert exposure.

The associated penalties are severe enough to make this exposure salient even where the probability of enforcement is modest. The International Emergency Economic Powers Act authorizes civil penalties up to the greater of a statutory maximum or twice the value of the offending transaction and, for willful violations, criminal penalties of up to $1,000,000 and twenty years' imprisonment.[1] The regime did, however, grant U.S. persons time-limited general licenses to wind down and divest covered positions within specified windows, which concentrated the practical pressure to exit covered relationships around the 2021 effective dates and avoided a diffuse temporal spread.

This role-, compensation-, and transaction-contingent exposure is the source of the individual-level legal and reputational risk that we later term compliance chill. The exposure is most acute for independent directors, who typically receive the fee- or equity-based compensation that falls within the regulations' reach and who, lacking an executive or controlling stake, bear lower switching costs than insider directors once the legal cost of continued service rises.

Beyond direct legal exposure, these executive orders, together with the broader sanctions framework directed at Chinese listed companies, likely created substantial constraints for U.S.-national directors even where no explicit prohibition applied. In practice, such directors may have found it increasingly difficult, and in some cases impractical, to continue serving on the boards of sanctioned or politically sensitive Chinese listed firms as legal, compliance, and reputational risks rose. The increasingly hostile geopolitical environment may also have encouraged foreign directors, as well as internationally connected directors based in China, to resign from Chinese boards even absent a direct legal prohibition, since the reputational and regulatory costs of such associations grew harder to justify.

3 Related Literature

This Article contributes to three distinct research streams: U.S.–China decoupling and its corporate consequences, board composition and director human capital, and the board-level mecha-

[1] 50 U.S.C. § 1705; the prohibitions and the "U.S. person" definition are codified at 31 C.F.R. part 586. The statutory civil maximum is adjusted periodically for inflation, and the precise regulatory cross-references should be confirmed against the OFAC regulations current at the time of submission.

nisms through which decoupling affects firm outcomes. We discuss each in turn before explaining how this Article links them.

3.1 U.S.–China Decoupling and Board Composition: Exit, Replacement, and Domestic Professionalization

Recent evidence suggests that director retention is sensitive to the geopolitical environment, because board service is embedded in cross-border legal, reputational, and identity contexts. As Tian et al. (2025) document, foreign directors are more likely to exit Chinese boards when bilateral political relations deteriorate. Complementing this identity-based mechanism, Liu et al. (2025) show that sanctions can generate "shock anticipation" in director labor markets, so that foreign directors may resign when their own firms are directly targeted and when sanctions imposed on peer firms raise perceived legal and reputational risk. This anticipatory resignation logic is consistent with the broader finding that outside directors tend to quit preemptively when they foresee reputational or legal exposure ahead (Fahlenbrach et al., 2010).

A board human capital perspective, however, requires moving beyond exit alone (Adams et al., 2010). Because board seats are typically refilled, the identity and expertise of replacement directors can meaningfully change a board's human capital and governance capacity. Indeed, a growing literature on China suggests that specific kinds of professional directors matter for governance. Zhu et al. (2016) show that more empowered independent directors engage in stronger monitoring and are associated with higher firm value, while Chen et al. (2019) show that forced resignations of academic independent directors in China reduce firm value, which suggests that academically trained directors contribute economically meaningful expertise. He and Liu (2016) show that Chinese listed firms appoint independent directors with professional legal backgrounds for specific consulting and governance purposes, thereby treating independent directors as heterogeneous sources of expertise. More broadly, Firth et al. (2016) show that regulatory sanctions imposed on independent directors in China have persistent consequences for their future board opportunities, which reinforces the plausibility of systematic director-labor-market adjustment after external shocks.

Regulation can also change board composition directly by determining which types of directors may serve. Fan (2021) studies China's regulation banning politicians from corporate boards and shows that this external regulatory shock led to substantial board reconfiguration and affected firm performance. Berkowitz et al. (2022) extend this logic, showing that firms that lost politically connected directors initially suffered a valuation decline but subsequently became more productive, transparent, and innovative, a pattern of short-run disruption followed by longer-run governance improvement. Both papers demonstrate that external political and regulatory shocks can reshape the mix of directors on boards, and that the consequences depend critically on who replaces those who depart. Building on these insights, this Article examines how U.S.–China decoupling pressures drive the exit of U.S.-nexus directors from Chinese listed firms, who replaces them, and what that substitution implies for overall board human capital. In particular, we examine whether firms respond by substituting toward domestically credentialed professional directors when U.S.-linked board human capital becomes more costly to maintain.

3.2 Board Composition and Foreign-Director Human Capital

A large literature studies how board composition affects firm governance and firm value (Adams et al., 2010). The board internationalization literature argues that foreign directors affect firm outcomes through both monitoring and advising channels, including the transfer of expertise and the reduction of information frictions (Masulis et al., 2012; Oxelheim and Randøy, 2003; Oxelheim et al., 2013). Related work suggests that foreign independent directors may be especially valuable where domestic legal institutions are weaker, or where the directors themselves come from stronger legal environments (Miletkov et al., 2017). Evidence from China similarly suggests that directors with overseas experience improve firm performance and the information environment (Giannetti et al., 2015), that foreign experience is associated with lower stock-price crash risk (Cao et al., 2019), and that it is associated with different payout policies (Tao et al., 2022). A related external-certification view argues that foreign-connected directors can enhance credibility with global investors and transaction partners, especially where liability-of-origin concerns are salient (Yu et al., 2024). More broadly, the cross-listing literature establishes the general principle that firms with tighter links to developed-market capital and governance standards command a valuation premium, because those links signal commitment to outside investors (Doidge et al., 2004). Because the departure of U.S.-connected directors in our setting is driven by an external political shock and is therefore separated from endogenous board choices, we can study how boards recompose when this internationally connected human capital becomes costly to retain, thereby distinguishing the exit from firms' ordinary board-selection decisions.

The Chinese governance literature also suggests that director quality matters for monitoring and value. As Liu et al. (2015) show, board independence has a positive overall effect on firm performance in China; Jiang et al. (2016) document that independent directors' reputation concerns shape their voting behavior; and Gong et al. (2021) find that stronger board independence can reduce tunneling by controlling shareholders. The tunneling channel is especially consequential in the Chinese context, where controlling shareholders have historically used intercorporate loans and related-party transactions to siphon resources from listed companies at the expense of minority shareholders (Jiang et al., 2010).

3.3 Decoupling, Firm Outcomes, and the Board Mechanism

A decoupling literature documents that U.S.–China tensions have real consequences for firm outcomes, although it generally does so without opening the board mechanism. Event-study evidence from the U.S.–China trade war documents broad stock-market reactions and spillovers (Egger and Zhu, 2020; He et al., 2022; Oh and Kim, 2024). Research on sanctions and export controls suggests that political restrictions affect cross-border investment, innovation, and strategic adjustment (Biglaiser and Lektzian, 2011; Hu et al., 2024; Liu and Li, 2023). While these studies establish that decoupling pressures have real and valuation consequences, they do not identify board reconfiguration as the mechanism through which these effects operate.

This Article contributes by linking these literatures. Existing board-composition studies often face endogeneity, because boards are chosen by firms and directors self-select into board positions, whereas existing decoupling studies document firm-level outcomes but rarely trace how geopolitical shocks reshape boards. By examining the exit of U.S.-nexus directors under U.S.–China decoupling pressure, the replacement process, and the resulting change in board human capital, this Article identifies board composition as a channel through which geopolitical shocks reshape the internal organization of Chinese listed firms, and documents the pattern of domestic substitution that follows.

4 Measuring US-Nexus Board Connections

Our empirical exercise requires an attribute that standard corporate-governance data do not report directly, namely whether an individual director holds a meaningful connection to the United States. We construct this attribute through a measurement procedure whose primary layer rests on auditable structured data, supplemented by a classification layer for the cases that structured fields cannot resolve.

Our primary measure classifies a director as U.S.-nexus if any of three conditions holds in CSMAR structured records: (i) U.S. nationality or permanent residency, (ii) at least five years of U.S. work experience, or (iii) a U.S. professional license. We apply this definition to the resume, education, professional-title, nationality, and licensure fields of the CSMAR Corporate Governance module. Because this structured measure is fully auditable, it does not depend on any model-based inference, and we accordingly treat it as the regressor underlying all of our headline estimates.

For directors whose U.S. connection we cannot verify from structured fields, we supplement the structured measure with biographical text retrieved from public web sources (i.e., Baidu search snippets), which a large-language model then classifies into a nationality / U.S.-nexus label. This procedure yields the market-specific classified files that we use in our robustness analysis. We treat the classification layer as supplemental, since it defines the expanded treatment definitions (v7 and v8 in the Appendix) and leaves the primary measure unchanged. As the Appendix shows, the headline exit estimate attenuates only mildly once we add these classified directors (from -0.020 to -0.018), which confirms that the result survives the addition of model-based labels. We therefore present the auditable three-prong construct as our measurement contribution and the classification layer as a robustness extension.

In the Hong Kong market, where structured nationality coverage is thinner, we infer director origin from a multi-dialect Chinese surname classifier (i.e., a classifier spanning Mandarin, Cantonese, Hokkien, and Hakka) that flags non-Chinese names. We validate this surname measure against an independent CSMAR-sourced nationality classifier, which covers 70.5% of Hong Kong director-years. The two instruments appear to capture largely distinct director populations, since the surname-based and nationality-based treatment groups overlap on only 25 firms, and neither treatment predicts the other's dependent variable (cross-DV checks, not significant at the 10% level). We read this pattern as triangulation, because the classifiers measure distinct populations and thereby serve as mutual cross-checks. The surname classifier is admittedly noisy, since it cannot distinguish a Hong Kong permanent resident bearing a Western surname from a U.S. citizen, and it misclassifies ethnically Chinese directors who hold Western citizenship. We therefore treat the surname measure as a broad indicator of Western-aligned directors and the nationality measure as the narrower, U.S.-specific instrument; in Section 8 we exploit the gap between the two to separate the U.S.-specific channel from broader de-internationalization.

5 Data and Empirical Strategy

5.1 Data and Sources

We procure our director and resume data from two CSMAR modules: the Corporate Governance module (annual snapshots 2010–2024, 176K unique directors) and the director changes file (event-level, $\sim$1M rows). The annual file supplies resume text, education background, professional titles, and overseas work and education indicators, while the event file records appointment and departure dates alongside resignation-reason text; from these fields we construct the U.S.-nexus measure described in Section 4.

The remaining firm-level variables come from three further CSMAR modules. We draw our financial controls—log assets, leverage (i.e., total liabilities over total assets), and ROA (i.e., net income over total assets)—from the Financial Statement modules, where they serve as time-varying covariates winsorized at the 1st and 99th percentiles. The Ownership and Control module supplies a time-varying state-owned/private (SOE/POE) classification together with top-10 shareholder concentration, while board and compensation variables—board size (i.e., the board-of-directors count), the independent-director ratio, top-three executive pay, and aggregate institutional ownership—come from the Executive Compensation and Governance module, with coverage exceeding 99% over the sample period.

Our estimation sample comprises 2,941 unique firms (1,109 treated and 1,832 control) observed over 2019–2024, yielding 17,171 firm-year observations. We classify a firm as treated if it had at least one U.S.-nexus director in the 2019–2020 baseline; control firms are matched by industry (CSRC two-digit) and internationalization status, and the post-shock indicator equals one for years from 2021 onward.

5.2 Estimating Equations

To examine how the decoupling shock reshapes board composition, we estimate an exposure-based difference-in-differences specification with firm and year fixed effects. For firm j in year t,

$$\text{U.S.-nexus share}_{jt} = \beta_1(\text{Treated}_j \times \text{Post}_t) + \gamma X_{jt} + \delta_j + \lambda_t + \epsilon_{jt}, \tag{1}$$

where Treated_j equals 1 for firms with at least one U.S.-nexus director in the 2019–2020 baseline, Post_t equals 1 for years ≥ 2021, X_{jt} collects log assets, leverage, and ROA, while δ_j and λ_t represent firm and year fixed effects, respectively, and standard errors are clustered at the firm level.

We expect the coefficient of interest, β_1, to be negative. A dose-response variant replaces Treated_j with the continuous pre-shock intensity $\overline{\text{U.S.-nexus share}}_{j,2019\text{–}2020}$, while an event-study variant replaces $\text{Treated}_j \times \text{Post}_t$ with a full set of year-by-treatment interactions.

We estimate the replacement analysis on matched exit–entry pairs, regressing an indicator for domestic-technocrat replacement on Treated U.S. exit_{ijt} (i.e., a U.S.-nexus director leaving a treated firm post-shock), again with firm and year fixed effects. As Section 7 makes explicit, this analysis is descriptive: it characterizes who fills vacated seats and does not isolate a causal effect of director loss on replacement type. The Hong Kong analyses adopt the same DiD structure, with treatment defined by baseline Western-surname or U.S.-nationality directors, whereas the explicit-sanctions analysis uses a binary indicator for the eleven sanctioned firms.

5.3 Identification Strategy

Our identification strategy follows the exposure-based difference-in-differences framework, under which pre-period exposure defines treatment and an external shock supplies the source of quasi-exogenous variation. We define treatment by whether a firm had at least one U.S.-nexus director during the 2019–2020 baseline period, and we aim to identify the effect by exploiting the sanctions shock—driven by U.S. national-security considerations quasi-exogenous to individual firm board composition—as a shift in the costs facing U.S.-connected directors. This structure parallels the "China shock" literature, in which Autor, Dorn, and Hanson (2013) define treatment intensity by pre-period import exposure and locate identification in the quasi-exogeneity of the trade shock. Goldsmith-Pinkham, Sorkin, and Swift (2020) and Borusyak, Hull, and Jaravel (2022) provide formal treatments of identification in shift-share designs. Our setting satisfies the "quasi-exogenous shock" condition because the timing, scope, and targeting of U.S. sanctions were determined by Washington's national-security apparatus, independent of the board composition of individual Chinese firms. Identification thus runs through the quasi-exogeneity of the shock. The shares themselves are plausibly endogenous to firm characteristics, and the firm and year fixed effects absorb time-invariant share-level confounders. Because the design rests on a single sanctions episode, we borrow the conceptual exposure-DiD logic of the China-shock literature while leaving its large-number-of-shocks asymptotic inference to settings with many independent shocks. This exposure-based design is aimed at the effect of the shock on the measured composition of the board—the outcome to which firms' U.S.-nexus exposure is mechanically linked—and we accordingly limit causal claims to that outcome, treating them as conditional on the identifying assumptions set out below.

The decoupling pressure we study operated as a regime rather than as a single instrument, since it was delivered through several legal instruments that fired in close succession: Executive Order 13959 (November 2020) and its broadening successor Executive Order 14032 (June 2021), the construction and expansion of the NS-CMIC list, the U.S. Persons Rule, and a contemporaneous tightening of the BIS Entity List. Because these instruments overlap both in timing and in the firms they reach, we estimate the cumulative response to the bundle; the data cannot separate the marginal effect of Executive Order 14032 from that of the Persons Rule or the Entity List when all bind within a twelve-month window. We identify the cumulative reduced-form effect of the decoupling regime as a whole, dated to its operative break on August 2, 2021, when the principal investment prohibitions took effect. For the policy question that motivates this Article—how geopolitical fragmentation reshapes corporate boards—the bundle is the relevant treatment, because U.S.-connected directors faced the entire regime. We treat the earlier 2018 trade-war measures, namely the Section 301 tariffs and the first Entity List designations, as a distinct prior shock that the 2019–2024 estimation window deliberately excludes, so that the 2021 break is not conflated with trade-war effects.

Because we define treatment by the level of the outcome variable in the pre-period, treated and control firms cannot exhibit parallel trends in that variable by construction, since treated firms were accumulating U.S.-nexus directors throughout 2015–2019 while control firms were not. The relevant identifying assumption therefore concerns the quasi-exogeneity of the sanctions shock to pre-period board composition: to establish the causal impact of the decoupling regime, the effect of this shock on board composition must arise solely through its influence on the costs facing U.S.-connected directors and be unrelated to any prior process by which Washington timed or targeted sanctions on the basis of which Chinese firms had Western-connected directors. The principal threat to this design is mean reversion. Because we select firms on a high pre-period U.S.-nexus level, a residual concern is that such firms might mechanically decline toward a common mean regardless of the shock; pre-trend parallelism is unattainable under this treatment definition and therefore cannot serve as the operative assumption. Under a counterfactual scenario where mean reversion drives the post-shock decline, one would expect the U.S.-nexus director share at high-baseline firms to fall back toward the average even absent any sanctions, and one would expect that decline to be shared by every category of director at those firms. Our central defense against this rival reading is a within-firm placebo. At the same treated firms, the post-shock decline is confined to U.S.-connected directors, whereas non-U.S. foreign directors are unaffected on the mainland and recruited in Hong Kong (Table 11). Mean reversion, mechanical turnover at the end of fixed board terms, and a market-wide de-internationalization trend would each depress all director types at a high-baseline firm, and therefore cannot generate the asymmetry between U.S. and non-U.S. directors that we observe within the same firm-years.

The triple-difference in the resignation analysis (U.S.-nexus $\times$ Post $\times$ Treated $= -0.122^{**}$) embeds this within-firm comparison directly. Three further patterns reinforce the reading. First, the U.S.-nexus director share shows no detectable divergence in the single available pre-treatment year, since the 2020 coefficient, $+0.0003$, is not significant at the 10% level relative to the 2019 base year. Second, post-shock exit is proportional to baseline exposure (dose-response coefficient $\beta = -0.369$, significant at the 1% level), which is consistent with a shock-driven mechanism because a common mean-reversion process would not naturally generate a steeper decline among firms with greater baseline exposure. Third, the resignation channel shifts discretely from voluntary ("personal reasons") to passive ("term expiry") after 2021, inconsistent with smooth compositional adjustment.[2] Our primary defense against mean reversion, however, is the director-level within-board exit-hazard design with firm $\times$ year fixed effects, which compares each U.S.-connected director only with the non-U.S. colleagues serving on the same board in the same

[2]For the same reason—treatment defined by pre-period outcome levels—synthetic control methods (Abadie 2021) and synthetic difference-in-differences (Arkhangelsky et al. 2021) are inapplicable, as the reweighting procedure cannot match treated firms' pre-treatment trajectories from control units with near-zero baseline exposure.

year and so differences out any firm-level trajectory by construction; we develop it in the Discussion (Figure 3; Table A4). A pre-shock pseudo-treatment defined on a 2014–2015 baseline, which would otherwise be the most direct remaining check, is infeasible in a panel that begins in 2018, and the within-board design supersedes it by removing the very firm-level selection the pseudo-treatment was meant to detect.

We restrict the estimation sample to 2019–2024 to isolate the primary sanctions escalation beginning in 2021 from earlier trade-war measures. The U.S.–China trade conflict escalated in stages: initial Entity List designations affected approximately 140 firms from 2018, followed by a relative pause in 2019–2020, and then the major sanctions escalation via Executive Order 14032 in June 2021 and subsequent expansions. Including 2015–2018 in the pre-period would conflate two distinct policy shocks—the early trade-war skirmishes and the later sanctions regime—and so would complicate identification of the 2021 break. The 2019–2024 window provides a brief pre-period during which the early trade-war effects had settled and the major sanctions had not yet taken effect. Shock-timing clarity supplies the rationale for this restriction, while parallel-trends concerns play a different role because the design relies on shock quasi-exogeneity as the relevant identifying assumption.

The Hong Kong cross-market extension provides additional support. In the extended event study (2015–2025), coefficients on the treated $\times$ year interaction are statistically insignificant for 2017, 2018, and 2019—three consecutive years of clean parallel trends before the 2021 break. This stronger pre-trend performance in Hong Kong plausibly reflects the later onset of board internationalization among Hong Kong-listed firms relative to mainland A-share companies, which yields a more favorable pre-treatment window. Figures 1 and 2 plot the mainland and Hong Kong event studies, respectively.

Our causal interpretation is bounded by where the design's identifying power lies, which is strongest for board composition, the outcome to which firms' pre-period U.S.-nexus exposure is mechanically and contemporaneously linked. We therefore treat the director-exit results, and the placebo decomposition that isolates the U.S.-specific component from a concurrent market-wide de-internationalization, as our most defensible causal interpretation, while acknowledging the residual threats to identification set out below. Downstream firm-level outcomes such as valuation and related-party activity lie further from the source of variation, since any effect of the shock on them would operate through multiple channels—direct sanctions exposure, customer and supplier disruption, financing conditions, and investor sentiment—that the exposure measure does not separate. We accordingly report the replacement and compensation patterns as descriptive associations conditional on the identified exit, leaving the causal effect of director loss on those outcomes unestimated. Appendix Figure A1 summarizes this identification logic and the underlying mechanism.

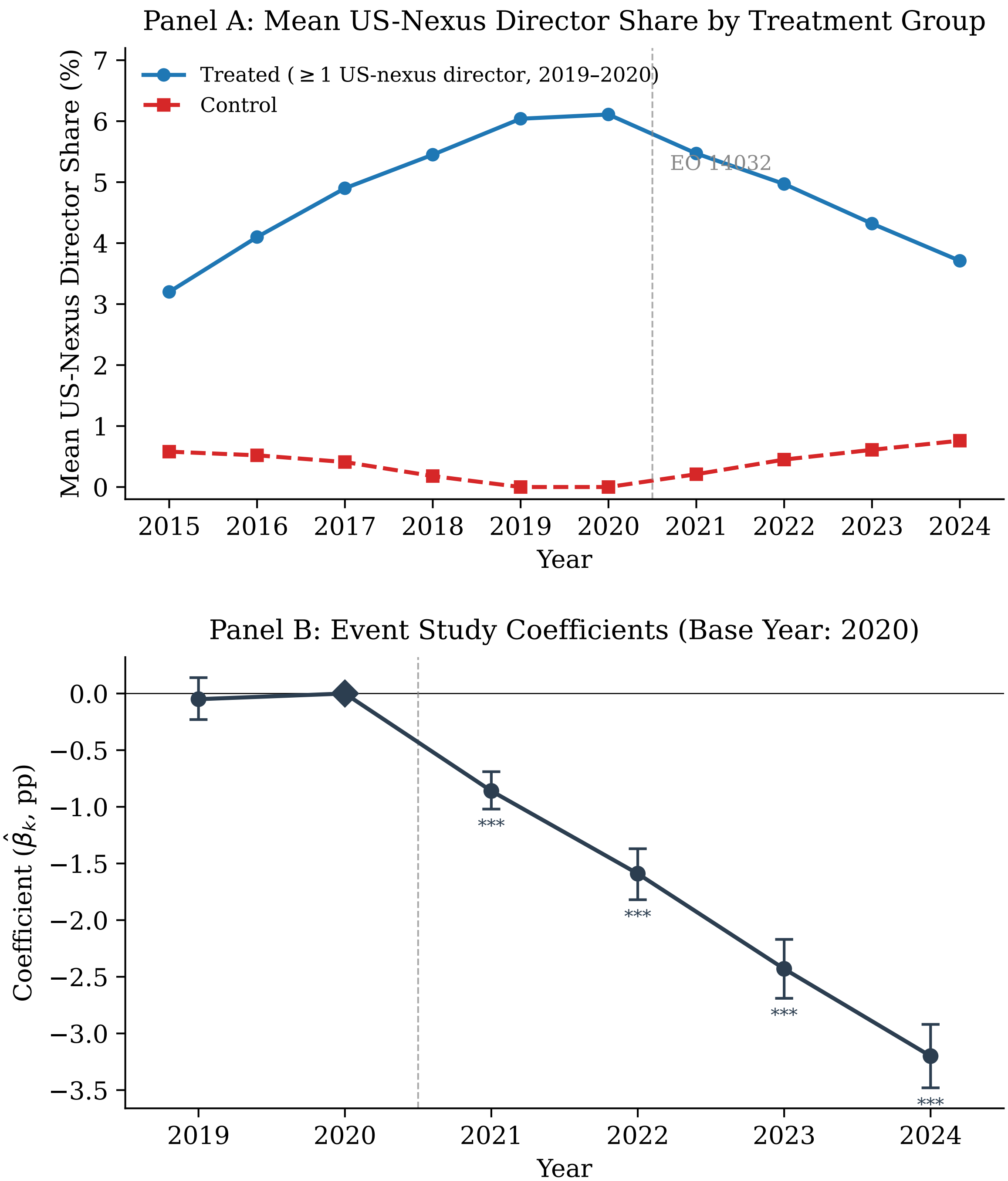

Figure 1: Event-study pre-trends, mainland A-share firms. Panel A plots the mean U.S.-nexus director share for treated ($\geq$ 1 U.S.-nexus director in 2019–2020) versus control firms; Panel B plots the year-by-treatment event-study coefficients (base year 2020) with 95% confidence intervals. The single available pre-treatment coefficient (2019, not significant at the 10% level) shows no detectable divergence, and the monotonic post-2021 decline is consistent with a shock-driven exit; identification rests on the quasi-exogeneity of the shock rather than on pre-trend parallelism, which a one-year pre-window cannot establish. Exact coefficients appear in Table 3.

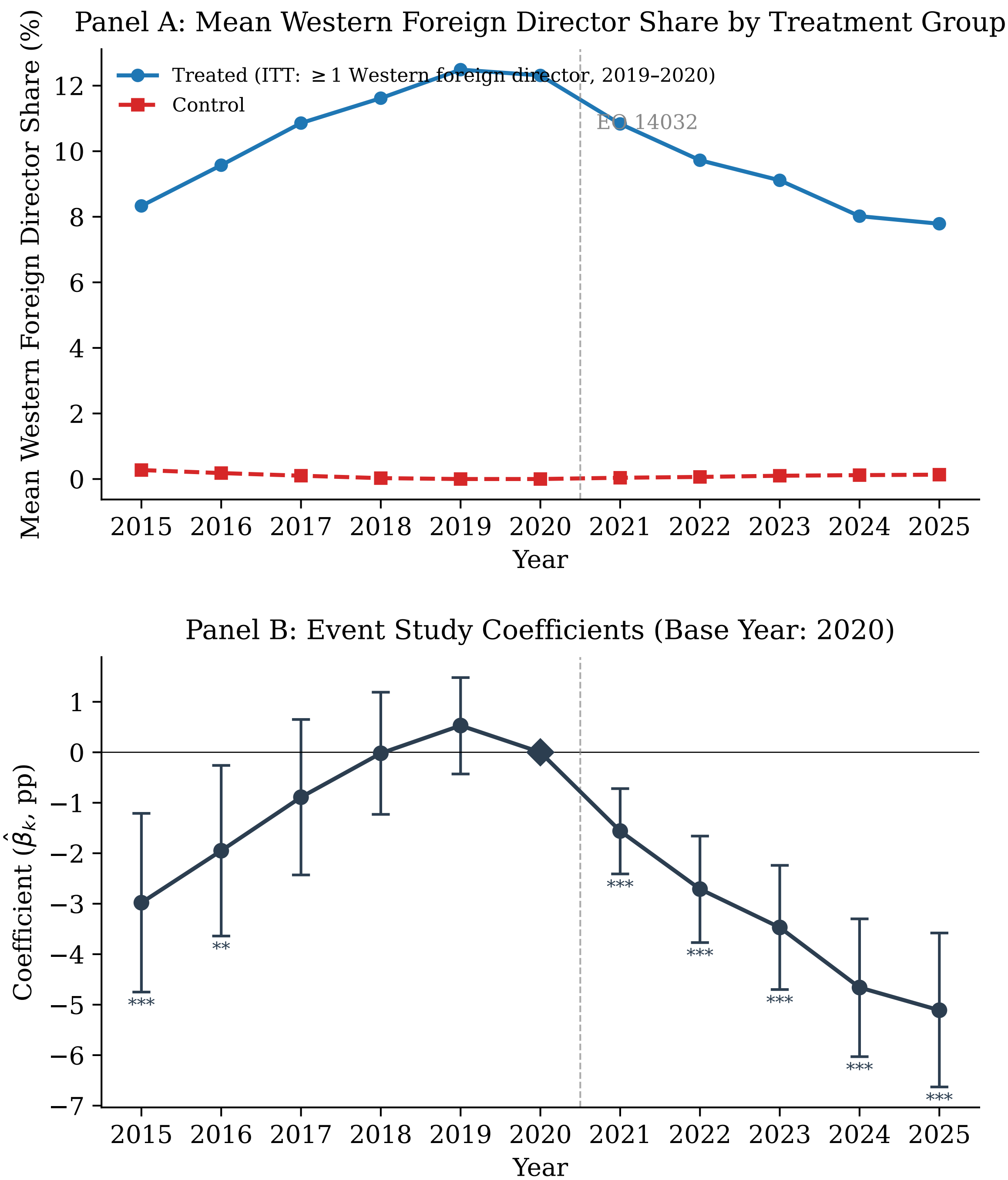

Figure 2: Event-study pre-trends, Hong Kong-listed Chinese firms. Panel A plots the mean Western-foreign director share for treated versus control firms; Panel B plots the year-by-treatment event-study coefficients (base year 2020) with 95% confidence intervals. The 2017–2019 pre-treatment window is clean (all not significant at the 10% level); the larger 2015–2016 coefficients reflect the onset of board internationalization in the mid-2010s rather than a sanctions response. The post-2021 decline mirrors the mainland pattern.

5.4 Summary Statistics

Table 1: Summary Statistics

Panel A: Firm-Year Variables (2019–2024)						
	N	Mean	SD	P25	Median	P75
U.S.-nexus director share	17,171	0.021	0.038	0.000	0.000	0.044
U.S.-nexus directors (count)	17,171	0.403	0.713	0.000	0.000	1.000
Foreign-director share	17,171	0.018	0.055	0.000	0.000	0.000
Board size	17,110	8.41	1.78	7.0	9.0	9.0
Independent-director ratio (%)	17,109	38.13	5.72	33.33	36.36	42.86
Log assets	17,171	22.73	1.631	21.62	22.47	23.56
Leverage	17,171	0.462	0.272	0.287	0.444	0.606
ROA	17,171	0.019	0.157	0.006	0.030	0.063
Top-3 executive pay (M¥)	17,088	4.32	4.56	2.09	3.08	4.93
Institutional ownership (%)	17,088	44.49	24.78	24.83	45.27	64.21
State-owned enterprise	17,171	0.367	—	—	—	—
Panel B: Treatment and Control Breakdown						
	Treated (1,109 firms)			Control (1,832 firms)		
	N	Mean	SD	N	Mean	SD
U.S.-nexus director share	6,433	0.053	0.044	10,738	0.002	0.010
Board size	6,429	8.61	1.89	10,681	8.29	1.70
Independent-director ratio (%)	6,432	38.10	5.78	10,677	38.14	5.69
Top-3 executive pay (M¥)	6,419	4.78	5.35	10,669	4.04	3.98
Institutional ownership (%)	6,427	44.89	24.77	10,661	44.25	24.79
Panel C: Sample Composition						
2,941 unique firms (1,109 treated, 1,832 control); 17,171 firm-years						
Pre (2019–2020): 5,679 firm-years; Post (2021–2024): 11,492 firm-years						
SOE firms: ∼1,026 (6,306 firm-years); POE firms: ∼2,149 (13,502 firm-years)						

Note: This table reports firm-year summary statistics for the preferred mainland A-share sample over 2019–2024 and the associated treatment-control breakdown.

Table 1 presents our summary statistics for the preferred sample of 17,171 firm-year observations spanning 2019–2024. Panel A reports firm-year-level variables. The average board has 8.4 directors (board of directors only, excluding the supervisory board and executives), with U.S.-nexus directors comprising 2.1% of the board on average. The mean independent-director ratio is 38.1%, just above the regulatory minimum of one-third, with 44% of firm-years bunched at exactly 33.3%. Top-3 executive compensation averages 4.32 million yuan, and aggregate institutional ownership is 44.5%, while state-owned enterprises comprise 36.7% of firm-years. Panel B breaks down key variables by treatment status: treated firms have higher baseline U.S.-nexus shares (5.3% vs. 0.2%), slightly larger boards, and higher executive compensation.

6 The Exit of US-Nexus Directors

We read the broad exit of U.S.-connected directors documented in this section—which appears across firms facing no direct legal prohibition, concentrates among independent directors, and operates through passive non-renewal at term end—as consistent with compliance chill, the channel through which the U.S. Persons Rule and the surrounding sanctions framework raised the personal legal and reputational cost of board service at politically exposed Chinese firms. Each of these three features maps onto the legal structure set out in Section 2. The concentration among independent directors follows from their fee- and equity-based compensation falling within the regulations' reach and from their lower switching costs; the appearance of the exit even at firms under no firm-specific prohibition follows from liability attaching personally to the U.S. person; and the clustering of departures at scheduled term end, with mid-term resignation playing little role, is the low-friction route by which a board sheds exposure once the legal cost of continued service rises. Note that the design identifies the director-exit outcome while leaving the legal-liability channel indirectly observed. We therefore present compliance chill as the reading most consistent with the assembled evidence, while acknowledging that the data do not adjudicate it against alternatives such as reputational avoidance unrelated to legal liability or anticipatory firm-side board management.

Table 2: Difference-in-Differences: US-Nexus Director Share

	(1) Preferred 2019–2024 Post≥2021	(2) Original 2018–2024 Post≥2022	(3) Early Post 2019–2024 Post≥2020	(4) Extended 2018–2024 Post≥2021
DV: U.S.-nexus director share				
Treated × Post	-0.0200^{***}	-0.0200^{***}	-0.0156^{***}	-0.0176^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Alternative DVs (Column 1 specification only)				
DV: U.S.-nexus directors (count)		$\hat{\beta} = -0.3692^{***}$, $p < 0.001$		
DV: Any U.S.-nexus director (0/1)		$\hat{\beta} = -0.3386^{***}$, $p < 0.001$		
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Observations	17,171	19,806	17,171	19,806
R^2 (within)	0.0504	0.0494	0.0183	0.0374

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The main panel estimates difference-in-differences effects on the U.S.-nexus director share across sample windows, with controls, firm fixed effects, year fixed effects, and firm-clustered inference.

Table 2 reports our base difference-in-differences estimates of the geopolitical shock on U.S.-nexus board representation. Our preferred specification (Column 1, the 2019–2024 sample with Post $\geq$ 2021) yields $\hat{\beta} = -0.020$ (significant at the 1% level), which indicates that treated firms experienced a 2.0 percentage-point decline in U.S.-nexus director share relative to control firms. Read against a treated-firm pre-shock mean of approximately 5.9%, this amounts to a one-third reduction. The estimate is notably stable across sample windows: extending the panel to 2018–2024 (Column 2, which also shifts the post cutoff to 2022) yields an identical point estimate, while holding the 2021 cutoff fixed over that window leaves it nearly unchanged (Column 4, -0.018***), and moving the post cutoff earlier to 2020 (Column 3) produces a somewhat smaller but still highly significant effect (-0.016***). Under industry × year fixed effects the estimate is virtually unchanged (-0.020***), which confirms that the result does not reflect sector-wide trends. Even under two-way firm and year clustering, the most demanding standard-error specification we report, the coefficient remains significant at the 1% level despite a fourfold increase in the standard error.

Table 3: Event Study: Year-by-Treatment Interactions

	(1) Full Sample 2018–2024 Base = 2020	(2) Preferred 2019–2024 Base = 2019
DV: U.S.-nexus director share		
Treated × 2018	-0.0075^{***}	
	(<0.001)	
Treated × 2019	-0.0004	
	(0.719)	
Treated × 2020		$+0.0003$
		(0.763)
Treated × 2021	-0.0086^{***}	-0.0083^{***}
	(<0.001)	(<0.001)
Treated × 2022	-0.0160^{***}	-0.0158^{***}
	(<0.001)	(<0.001)
Treated × 2023	-0.0244^{***}	-0.0242^{***}
	(<0.001)	(<0.001)
Treated × 2024	-0.0321^{***}	-0.0319^{***}
	(<0.001)	(<0.001)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Clustering	Firm	Firm
Observations	19,806	17,171

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports year-by-treatment interactions for the U.S.-nexus director share, with base years and sample windows shown in the column headers.

Table 3 presents our event-study coefficients. In the preferred 2019–2024 specification (Column 2, with base year 2019), the single available pre-treatment coefficient (2020) is $+0.0003$ (not significant at the 10% level), showing no detectable pre-shock divergence in the one pre-treatment year the window affords. The post-shock coefficients decline monotonically thereafter: -0.008^{***} (2021), -0.016^{***} (2022), -0.024^{***} (2023), and -0.032^{***} (2024). The full 2018–2024 specification (Column 1, with base year 2020) reveals a significant 2018 coefficient (-0.0075^{***}) alongside a clean 2019 reading (-0.0004, not significant at the 10% level). We regard the 2018 anomaly as an isolated blip attributable to the U.S.–China trade war, since it is smaller than every post-treatment coefficient and, as we show in the Discussion, is driven by a small set of early-treated firms and disappears once they are excluded. We accordingly adopt the 2019–2024 window as our primary specification.

Table 4: Dose-Response and Position-Type Concentration

Panel A: Extension 1a — Dose-Response by Position Type

	(1) U.S.-nexus share (independent)	(2) U.S.-nexus share (non-independent)
Intensity × Post	-0.6966^{***}	-0.1173^{***}
	(<0.001)	(<0.001)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Clustering	Firm	Firm
Observations	17,171	17,171

Panel B: Extension 1c — Position Type Split (Binary Treatment)

	(1) U.S.-nexus share (independent)	(2) U.S.-nexus share (non-independent)	(3) Independent-director share
Treated × Post	-0.0477^{***}	-0.0063^{***}	+0.0017
	(<0.001)	(<0.001)	(0.670)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustering	Firm	Firm	Firm
Observations	17,171	17,171	10,809
R^2 (within)	0.0503	0.0107	

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The panels compare dose-response and binary-treatment estimates by director position type; the Panel B independent-to-non-independent coefficient ratio is 7.5×.

Table 4 reports two extensions. Panel A presents the dose-response gradient by position type, which is 6× steeper for independent directors (-0.697^{***}) than for non-independent directors (-0.117^{***}). Panel B reports the binary difference-in-differences by position type: -0.048^{***} for independent directors against -0.006^{***} for non-independent directors, a 7.5× ratio. We read this concentration among independents as consistent with the U.S. Persons Rule bearing on advisory relationships and with independent directors facing lower switching costs. Notably, the board independence ratio itself is unaffected ($+0.002$, not significant at the 10% level), which confirms that firms replaced departing U.S.-nexus independents with domestic independents and maintained board structure.

Table 5: Resignation Reasons as Quasi-exogeneity Evidence

Panel A: Euphemistic Resignation Rates (2×2)				
	Pre (2019–20)	Post (2021–24)	Diff	
U.S.-nexus	35.5% (n=403)	31.3% (n=838)	-4.2 pp	
Non-U.S.-nexus	38.0% (n=25,392)	38.6% (n=53,686)	$+0.6$ pp	
DiD $= -4.8$ pp, $\chi^2 = 22.03$, $p = 0.0001$				
Panel B: Linear probability model (DV: euphemistic resignation)				
	(1) Raw OLS	(2) Year FE	(3) Firm + Year FE	(4) Triple-Diff
U.S.-nexus × Post	-0.0666^{**}	-0.0652^{**}	-0.0701^{**}	
	(0.027)	(0.030)	(0.022)	
U.S.-nexus × Post × Treated				-0.1221^{**}
				(0.011)
U.S.-nexus × Post				$+0.0434$
				(0.347)
U.S.-nexus × Treated				-0.0974^{**}
				(0.035)
Year FE	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Controls	No	Yes	Yes	Yes
Observations	54,117	54,117	54,117	54,117

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports resignation-reason evidence using euphemistic resignation rates and linear probability models for U.S.-nexus departures before and after 2021.

Table 5 examines the stated reasons for resignation. Against the naive prediction that euphemistic reasons would increase, the post-shock U.S.-nexus departures show a decrease: $\hat{\beta}$(U.S.-nexus × Post) $= -0.065$ (significant at the 5% level). The triple interaction with Treated is stronger still (-0.122, significant at the 5% level). Decomposing the reasons reveals the mechanism, since "personal reasons" declined (-0.077, significant at the 1% level) while "term expiry" increased ($+0.061$, significant at the 10% level), and the effect concentrates among independent directors (-0.123, significant at the 1% level). Taken together, these patterns indicate that U.S.-nexus exits occurred through passive non-renewal at term end, with firms waiting for natural board renewal cycles and forced mid-term resignations playing little role, which is precisely the departure pattern a chilling-cost mechanism predicts.

7 Board Reconstitution: Who Replaces the Departing Directors

In this section we characterize who fills the seats vacated by U.S.-nexus directors. Having interpreted the exit as the likely causal effect of the shock—subject to the threats to identification discussed in Section 10—we now turn to how boards recompose themselves around it. We treat this second step as associational, because it conditions on the identified exit and describes the composition of replacements without asserting a separate causal effect, since replacement type is jointly determined with the firm's broader response to the shock.

Table 6: Replacement-Director Professionalization

Panel A: DV = Domestic technocrat (Replacement-Pair Level)				
	(1) Raw OLS	(2) Year FE	(3) Firm + Year FE	(4) Full
Treated (U.S.-nexus exit)	$+0.0282^{**}$ (0.014)	$+0.0411^{***}$ ($<$0.001)	$+0.0108^{**}$ (0.025)	$+0.0108^{**}$ (0.024)
Year FE	No	Yes	Yes	Yes
Firm FE	No	No	Yes	Yes
Controls	No	No	No	Yes
Observations	314,050	314,050	314,050	314,050
R^2	0.0001	0.0025		
Panel B: DV = CCP-affiliated replacement (Preferred Specification)				
	(1)			
Treated (U.S.-nexus exit)	-0.0012 (0.765)			
Year FE	Yes			
Firm FE	Yes			
Controls	Yes			
Observations	314,050			

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table uses replacement-pair data to estimate domestic technocrat and CCP replacement outcomes for seats vacated by U.S.-nexus exits.

Table 6 examines the directors who step into the seats that departing U.S.-nexus directors leave behind. Drawing on 314,050 matched exit–entry replacement pairs, our preferred specification (Column 4) yields $\hat{\beta}(\text{Treated U.S. exit}) = +0.011$ (significant at the 5% level), so that U.S.-nexus exits from treated firms post-shock are associated with a 1.1 percentage-point higher rate of replacement by domestic technocrats. We read this association as a descriptive pattern in who fills the vacated seats. The CCP-replacement rate, by contrast, exhibits no differential change

(-0.001, not significant at the 10% level), a clean descriptive non-result that runs directly against a state-capture narrative.

Table 7: Replacement: Credentials, Board Size, and Compensation

Panel A: Extension 2a — Technical-background index and Components

DV:	(1) Technical-background index	(2) Graduate degree	(3) Professional credential	(4) Domestic only	(5) Non-CCP
Treated (U.S.-nexus exit)	+0.0058**	+0.0052	+0.0143***	+0.0024	+0.0012
	(0.011)	(0.359)	(0.002)	(0.476)	(0.765)
Firm FE + Year FE + Controls	Yes	Yes	Yes	Yes	Yes
Observations	314,050	314,050	314,050	314,050	314,050

Panel B: Extension 2b — Board Size and Independence Ratio

	(1)	(2)	(3)
DV:	Board size	Independent-director ratio	Director count (incl. supervisors)
Source:	CSMAR comp/gov	CSMAR comp/gov	Director panel
Treated × Post	+0.0090	−0.0859	−0.3183**
	(0.821)	(0.611)	(0.013)
Firm FE + Year FE + Controls	Yes	Yes	Yes
Observations	17,110	17,109	17,171

Panel C: Extension 2c — Compensation

	(1)	(2)	(3)	(4)	(5)
DV:	Log(indep. director pay)	Log(indep. director pay)	Log(indep. director pay)	Log(top-3 exec. pay)	Log(top-3 exec. pay)
Sample:	Full	POE only	SOE only	Full	SOE only
Treated × Post	+0.1009	+0.1596**	−0.0364	−0.0142	−0.0574**
	(0.107)	(0.028)	(0.778)	(0.357)	(0.022)
Firm FE + Year FE + Controls	Yes	Yes	Yes	Yes	Yes
Observations	7,425	5,437	1,988	17,075	5,467

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. Board size counts only the board of directors (mean 8.4); the director count also includes supervisors and executives (mean 19.1), so the apparent shrinkage in that count is an artifact. Independent-director pay is measured at the director level; top-three executive pay is measured at the firm level.

Table 7 reports three extensions. Panel A shows that the composite technical-background index is higher for replacements of U.S.-nexus exits ($+0.006$, significant at the 5% level), a movement driven entirely by the professional-credential component ($+0.014$, significant at the 1% level). Panel B turns to board size: CSMAR's pre-computed board size (board of directors only, mean 8.4) registers no change ($+0.009$, not significant at the 10% level), consistent with complete one-for-one replacement. Notably, the previously reported shrinkage (-0.32, significant at the 5% level) was an artifact of the director-panel count, which erroneously included supervisory-board members and executives (mean 19.1). The CSMAR independent-director ratio is likewise unchanged (-0.086pp, not significant at the 10% level), which confirms that the independence ratio was maintained. Panel C examines compensation: independent-director pay at privately owned enterprises rose by 16% ($+0.160$, significant at the 5% level). We read this as an associational pattern consistent with upward pressure on independent-director pay following the exit of U.S.-connected directors, although it may also reflect a general post-shock repricing of director risk, and we do not interpret it as an identified causal effect. Top executive compensation is null overall (-0.014, not significant at the 10% level), while SOE executive pay declined by 5.7% (significant at the 5% level), a different labor market that is likely shaped by administrative pay constraints. The Appendix consolidates robustness across all specifications, including alternative fixed-effect structures, clustering levels, and wild cluster bootstrap inference (Table A2).

8 Cross-Market Evidence and the Compliance-Chill Threshold

The mainland results establish that the geopolitical shock caused U.S.-nexus directors to exit Chinese A-share-listed firms, and they document the domestic professionalization of the replacement directors that followed. A natural question is whether these effects generalize beyond the A-share universe and how the foreign investor base responded. The Hong Kong Stock Exchange hosts a distinct population of Chinese firms, many of them internationally oriented, which are subject to different regulatory and disclosure regimes and which report quarterly ownership breakdowns that are unavailable in A-share data. In this section we extend the analysis to this independent sample, test whether the same shock produced parallel director exits, and ask how substitution and capital reallocation differ across the two markets.

8.1 Foreign-Director Exit in Hong Kong

Table 8: Foreign-Director Exit: Hong Kong-Listed Chinese Firms

	(1)	(2)	(3)	(4)
	DV: Western-foreign director share		DV: Foreign-dialect director share	
Panel A: Baseline DiD				
Treated × Post	-0.0274^{***}	-0.0253^{***}	-0.0260^{***}	-0.0224^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Panel B: Dose-Response				
Intensity × Post	-0.2695^{***}	-0.2449^{***}	-0.1199^{***}	-0.1092^{***}
	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Controls	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Panel A statistics				
Observations	23,869	21,612	23,869	21,612
Firms	2,606	2,535	2,606	2,535
R^2 (within)	0.0450	0.0418	0.0214	0.0183
Panel B statistics				
R^2 (within)	0.0972	0.0884	0.0380	0.0349

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The panels report baseline difference-in-differences and dose-response estimates for Hong Kong-listed Chinese firms using Western-surname and foreign-dialect director-share outcomes.

Table 8 replicates the mainland exit result in an independent panel of Hong Kong-listed Chinese firms (roughly 2,600 firms over 2014–2025, with estimation samples that are somewhat smaller after controls and coverage restrictions, as reported in the table). We define treatment as having ≥ 1 Western-surname director in the 2019–2020 baseline, with Post $\geq$ 2021. Panel A reports the baseline DiD. The Western-surname measure yields $\hat{\beta} = -0.025$, an estimate that is significant at the 1% level, so that treated firms experienced a 2.5 pp decline in Western director share, closely paralleling the mainland 2.0 pp figure. The broader foreign-dialect measure produces a similar effect (-0.022, significant at the 1% level), and Panel B confirms a dose-response gradient (-0.245, significant at the 1% level).

An independent nationality-based classifier decomposes this effect. Defining treatment as ≥ 1 U.S.-nationality director in the 2019–2020 baseline (137 firms, excluding the 11 sanctioned) and using the share of U.S.-nationality directors as the dependent variable, we obtain an intent-to-treat estimate of -0.013, which is significant at the 5% level (Table 11). The gap between the surname estimate (-0.025) and the nationality estimate (-0.013) quantifies the non-U.S. Western component of the exit, since approximately half the surname-based effect reflects the departure of directors who are Western-aligned but not U.S. nationals, a pattern consistent with a broader de-internationalization trend distinct from U.S. sanctions. Because both rely on nationality-defined treatment, the nationality estimate provides the most direct comparison with the mainland figure.[3]

At intent-to-treat firms, the share of non-U.S. foreign directors rose post-shock ($+0.010$, significant at the 5% level), a movement consistent with boards offsetting U.S.-connected departures by recruiting other international directors, although the design cannot confirm that the recruitment was a deliberate substitution response. This pattern stands apart from the mainland one, in which firms replaced U.S.-nexus directors with domestic technocrats. The divergence plausibly reflects different labor-market depths, since the Hong Kong exchange hosts a deep pool of non-U.S. international professionals (the baseline non-U.S. foreign share is 5.6%), whereas mainland A-share boards draw from a domestic talent market.

At the eleven explicitly sanctioned firms, the pattern differs. Non-U.S. foreign directors also declined (-0.029, significant at the 10% level), even though these directors carried no direct U.S. compliance exposure. This marginal estimate, drawn from a small set of firms, is consistent with explicit sanctions generating a broader reputational deterrent that reaches foreign directors of all origins, though the small sample warrants caution.

[3] All Hong Kong director-exit results are robust to controlling for log market capitalization (22 of 22 specifications unchanged). We omit this control from the baseline to avoid observation loss (6.4% missing) and a potential bad-control concern, as market valuations may themselves respond to the shock.

8.2 International Ownership and the Capital-Base Response

Table 9: International-Ownership Decline: Explicitly Sanctioned Hong Kong Firms

	(1) International	(2) International	(3) Mainland	(4) Stock Connect
DV: Ownership share (%)				
Sanctioned × Post	-11.05^{***}	-10.36^{***}	$+3.04^{**}$	$+11.96^{***}$
	(<0.001)	(<0.001)	(0.020)	(0.004)
Controls	No	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm
Observations	79,495	71,510	71,509	20,120
Firms	2,398	2,307	2,307	739
R^2 (within)	−0.0003	0.0041	0.0005	0.1038

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table uses the Hong Kong quarterly ownership panel for explicitly sanctioned firms and reports ownership-share outcomes with firm and quarter fixed effects.

Table 9 tests whether explicitly sanctioned Hong Kong-listed firms experienced a decline in international institutional ownership. Treatment is binary, comprising the 11 firms designated on U.S. sanctions lists. The quarterly panel comprises 89,621 firm-quarter observations (2,398 firms) in the raw panel, with ownership breakdowns sourced from Wind Terminal, and estimation samples are smaller after controls and coverage restrictions (Table 9). Sanctioned firms experienced a 10.4 pp decline in international ownership (significant at the 1% level), accompanied by offsetting increases in mainland Chinese ownership ($+3.0$ pp, significant at the 5% level) and Stock Connect holdings ($+12.0$ pp, significant at the 1% level). The reallocation appears to operate through the northbound/southbound Connect mechanism, with direct mainland purchases playing little role. In the broader intent-to-treat sample, by contrast, ownership effects are uniformly null for international and Chinese ownership (Appendix Table A3; all not significant at the 10% level), with only a negative Stock Connect effect. Measurable financial-capital reallocation thus appears only among firms subject to legally binding investment prohibitions, although the intent-to-treat nulls partly reflect lower baseline foreign exposure and therefore provide limited evidence on investor non-response.

8.3 The Compliance-Chill Threshold

The cross-market evidence is consistent with a single mechanism operating at different intensities of legal exposure. Under compliance chill (i.e., intent-to-treat exposure with no binding prohibition), the response is selective: only U.S.-connected directors leave, while non-U.S. foreign directors are unaffected on the mainland and are recruited in Hong Kong. Under legal compulsion through explicit sanctions, the response becomes indiscriminate, as foreign directors of all origins decline and international capital reallocates. Table 10 lays the three regimes side by side.

Table 10: Selective Compliance Chill versus Indiscriminate Sanction: Cross-Market Board and Ownership Response

	Mainland (ITT)	HK (ITT)	HK (Sanctioned)
U.S./Western director exit	-0.020^{***}	-0.013^{**} (U.S.-nat.); -0.025^{***} (surname)	-0.014 (n.s., n=11)
Non-U.S. foreign response	$+0.001$ (n.s.)	$+0.010^{**}$	-0.029^{*}
Replacement type	Domestic technocrat	Non-U.S. international	Not directly measured (foreign share declines)
Intl. ownership response	Null (QFII ≈0.03% baseline)	Null; Stock Connect -2.47^{***}	Intl -10.4^{***}; CN $+3.0^{**}$; Connect $+12.0^{***}$
Selectivity	Selective	Selective (substitution)	Indiscriminate

Note: Cell entries are difference-in-differences coefficients on director-share or ownership-share outcomes; parenthetical notes give significance or classifier. Stars: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Treated firms: mainland 1,109; Hong Kong ITT 137 (U.S.-nationality); Hong Kong sanctioned 11. The sanctioned U.S.-nationality cell is imprecise ($n = 11$); the indiscriminate reading rests on the significant non-U.S. (-0.029^{*}) and international-ownership (-10.4pp) cells rather than that point estimate. Sources: Tables 2, 6, 7, 8, 9, 11, and A3.

The contrast yields the paper's central cross-market finding, namely that human capital and financial capital respond at different thresholds of geopolitical pressure. Director exit appears across the full intent-to-treat sample, whereas measurable ownership reallocation appears only where holding a specific security became legally prohibited. We caution that the ownership nulls partly reflect negligible baseline foreign exposure, since A-share sanctioned firms held only about 0.03% QFII pre-shock, which limits what they can reveal about investor response, and that "compliance chill" is our interpretation inferred from the director pattern.

9 Disentangling Sanctions from De-Internationalization

Table 11: Non-US Foreign-Director Response: Cross-Market Placebo Tests

	(1) Mainland	(2) HK (ITT)	(3) HK (Sanctioned)	(4) HK (Placebo T2)
Panel A: US/Western Director Exit (Actual Effect)				
Treated × Post	-0.0200^{***}	-0.0131^{**}	-0.0135	-0.0001
	(<0.001)	(0.012)	(0.309)	(0.894)
Treatment definition	U.S.-nexus	U.S.-nationality	Sanctioned	Other foreign
DV	U.S.-nexus share	U.S.-nationality share	U.S.-nationality share	U.S.-nationality share
Panel B: Non-US Foreign Directors at Treated Firms (Placebo)				
Treated × Post	$+0.0007$	$+0.0101^{**}$	-0.0294^{*}	-0.0236^{***}
	(0.484)	(0.034)	(0.053)	(<0.001)
DV	Non-U.S. foreign share	Non-U.S. foreign share	Non-U.S. foreign share	Non-U.S. foreign share
Treated firms	1,109	137	11	519
Control firms	1,841	2,515	2,652	1,595
Observations	17,171	21,566	21,612	19,301
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. Columns 1–3 define treatment by baseline U.S.-nexus, U.S.-nationality, or sanctions exposure; column 4 defines placebo treatment as at least one non-U.S. foreign director in the baseline and zero U.S. directors.

A central question for our identification strategy is whether the director exit we document reflects U.S.-specific sanctions pressure or a broader de-internationalization of corporate boards. To address it, we report placebo designs across both markets in Table 11, and the resulting pattern furnishes the paper's most credible secondary check on identification. Under a counterfactual scenario in which market-wide de-internationalization drives the exit, one would expect non-U.S. foreign directors to leave treated firms alongside their U.S.-connected colleagues; the placebos test that prediction directly.

The within-firm placebo in columns 1–3 holds the firm fixed. At mainland treated firms, non-U.S. foreign directors are entirely unaffected ($+0.001$, not significant at the 10% level), so that the exit is confined to the U.S.-nexus group. At Hong Kong intent-to-treat firms, by contrast, non-U.S. foreign directors increased ($+0.010$, significant at the 5% level), a movement consistent with substitution toward non-U.S. internationals. At explicitly sanctioned firms, non-U.S. foreign directors declined (-0.029, significant at the 10% level), which suggests that formal legal sanctions create a broader reputational deterrent reaching all foreign directors.

Column 4 reports an unexposed-firm placebo. Firms that held non-U.S. foreign directors but no U.S. directors in the baseline experienced a marked decline in their own-type directors (-0.024, significant at the 1% level), revealing a concurrent market-wide de-internationalization in Hong Kong boards that operates independently of U.S. sanctions. The event study displays clean pre-trends in 2018–2019 and a monotonic decline after 2021, a profile that points to the National Security Law, COVID-era border restrictions, and the wider Sino-Western tensions as contributors to foreign-director attrition across the market. An analogous test on the mainland, among firms with non-U.S. foreign directors but no U.S.-nexus exposure, likewise yields a significant decline (-0.016, significant at the 1% level; Table A1).

Taken together, the cross-market pattern supports three conclusions. First, the evidence points to a genuine U.S.-specific channel, because at treated firms U.S. directors exit while non-U.S. foreign directors are unaffected on the mainland and increase in Hong Kong, a pattern that diverges from the de-internationalization counterfactual. Second, a distinct de-internationalization channel operates on different firms and through a different mechanism. Third, the surname-based estimate (-0.025^{***}) captures both channels, whereas the nationality-based estimate (-0.013^{**}) isolates the U.S.-specific component, and the gap between the two is consistent with the market-wide trend we observe at unexposed firms. Appendix B reports robustness to alternative treatment definitions.

10 Discussion and Limitations

With the cross-market evidence in place, this section sets out the qualifications that bound its interpretation, addressing in sequence the threats to identification, the caveats attaching to our measures, and the extensions that would tighten the design. Our identification rests on the quasi-exogenous timing of the U.S. sanctions, namely the Entity List expansions, Executive Order 14032, and the U.S. Persons Rule. In our preferred 2019–2024 specification the parallel-trends requirement is satisfied cleanly, since the single pre-treatment coefficient (2020, base year 2019) is $+0.0003$ and is not significant at the 10% level. The extended 2018–2024 specification, by contrast, returns a 2018 coefficient that is negative and significant at the 1% level (-0.0075), a pattern we trace to 140 "early-treated" firms whose U.S.-nexus departures coincided with the Section 301 trade-war tariffs of July 2018; once these firms are excluded the 2018 anomaly disappears and the coefficient flips to $+0.003$. We therefore adopt the 2019–2024 window as our baseline. Should the Entity List addition dates generate genuinely staggered treatment timing, a Callaway–Sant'Anna (2021) heterogeneity-robust estimator would be the appropriate response, and we leave this extension for future implementation. The explicit Entity List data are too thin for standalone analysis—42 firms, 26 of them in the panel, only 7 overlapping with the treatment group—a scarcity that itself vindicates our intent-to-treat approach.

Because we define treatment by the pre-period level of the outcome, the residual concern is a mechanical-attrition account: the post-shock decline might reflect regression to the mean from a high-baseline group or routine non-renewal at the end of three-year board terms. We address this concern directly with a director-level exit-hazard design that holds firm and year jointly fixed. The unit of analysis is the director-year and the outcome is whether a director departs the board before the following year, while the specification absorbs firm $\times$ year fixed effects, so that the estimate compares each U.S.-connected director only with the non-U.S. colleagues serving on the same board in the same year. Any firm-level trajectory, including the mean-reverting drift of a high-baseline firm, is differenced out by construction, and a firm-level pre-trend therefore cannot produce the result. As Figure 3 shows, the within-board exit gap displays no positive pre-trend: the 2019 coefficient is small and not significant at the 10% level ($+1.6$pp), and the 2018 coefficient is significantly negative (-3.1pp), the opposite sign from the post-shock gap. The gap then rises monotonically once the sanctions bind, reaching $+2.0$pp in 2021, $+3.6$pp in 2022 (significant at the 1% level), and $+4.5$pp in 2023 (significant at the 1% level) against a mean annual exit rate near 14.5%. The pooled difference-in-differences is $+3.0$pp (significant at the 1% level) over the full 2018–2023 window and $+3.8$pp (significant at the 1% level) within treated firms, attenuating to $+1.5$pp (not significant at the 10% level) only in the narrow 2019–2023 window, where the growing effect is averaged against a near-zero 2020 base (Table A4). Scheduled term cycles cannot account for the gap, since routine turnover falls equally on a director's U.S. and non-U.S. board colleagues and is absorbed by the fixed effects. A pre-shock pseudo-treatment built on a 2014–2015 baseline is infeasible in the present panel, which begins in 2018; the within-board design supersedes that check, however, because it removes by construction the firm-level selection that the pseudo-treatment was meant to detect.

Because our treatment bundles several legal instruments together with other events of the 2020–2021 period—the pandemic, the broader trade conflict, and, in Hong Kong, the National Security Law—a natural concern is that one of these contemporaneous shocks produced the exit. The within-board design speaks to this concern as squarely as it does to mean reversion. Any shock that operates at the level of the firm, depressing all board service or all foreign board service at an exposed firm, is absorbed by the firm $\times$ year fixed effects and cannot generate the estimated gap. To survive the within-board comparison, a confounder would have to drive out a firm's U.S.-connected directors differentially relative to their own non-U.S. colleagues, and to do so on the timing of the 2021 sanctions. The pandemic, the trade war, and the National Security Law carry no such U.S.-specific, within-board signature, whereas the U.S. Persons Rule carries one by construction. The within-firm placebo reinforces this reading, since at mainland treated firms only U.S.-connected directors leave while non-U.S. foreign directors are unaffected (Table 11), which isolates the U.S.-specific component of the decoupling bundle from the concurrent market-wide de-internationalization.

The replacement analysis is descriptive, in the sense that it characterizes who fills vacated seats among matched exit–entry pairs while leaving the causal effect of director loss on replacement type unestimated. The domestic-technocrat result ($+0.011$, significant at the 5% level) is sensitive to how we define the CCP keyword set: a strict three-keyword definition weakens it to

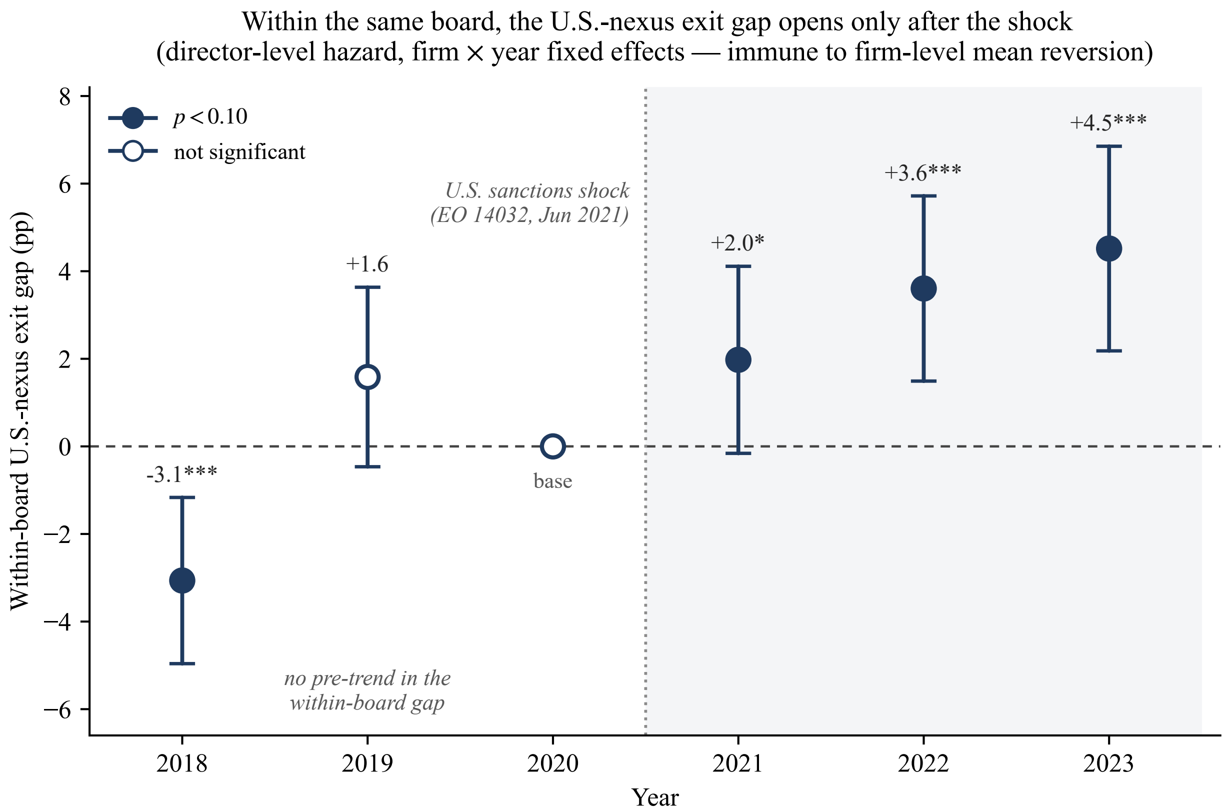

Figure 3: Within-board exit hazard for U.S.-nexus directors, by year. Each point is the coefficient on U.S.-nexus $\times$ year in a director-level linear-probability model of board exit with firm $\times$ year fixed effects (base year 2020), estimated on 323,098 director-years across 2,941 firms; whiskers are firm-clustered 95% confidence intervals (filled markers denote $p < 0.10$). Because the firm $\times$ year fixed effects absorb every firm-level confounder, the coefficient compares U.S.-connected directors only with their non-U.S. colleagues on the same board in the same year, so a firm-level pre-trend or mean-reverting drift cannot generate it. The within-board gap shows no positive pre-trend (2019, $+1.6$pp, not significant at the 10% level; 2018 significantly negative, -3.1pp) and climbs monotonically after the 2021 sanctions to $+4.5$pp by 2023 (significant at the 1% level), against a mean annual exit rate near 14.5%. Full estimates appear in Table A4.

a coefficient that is not significant at the 10% level, while a broad ten-keyword definition flips the sign (-0.006, not significant at the 10% level), because the domestic-technocrat indicator is a residual category and the boundary between party-affiliated and party-adjacent careers is inherently ambiguous. Because the replacement-pair matching uses annual data, exact exit–entry pairing is only approximate. The board-size correction nonetheless stands. Using CSMAR's precomputed board-size measure, the coefficient is null ($+0.009$, not significant at the 10% level), which confirms complete one-for-one replacement and reveals the earlier -0.32-seat shrinkage to have been an artifact of a director-panel count that included supervisory and executive positions. Note that the replacement result also weakens to a coefficient that is not significant at the 10% level when we exclude the 140 early-treated firms ($+0.004$), consistent with those firms driving part of the effect.

The mainland arm finds no institutional-ownership shift at treated A-share firms. Three data sources independently produce null aggregate results: a top-10 shareholder origin decomposition, aggregate institutional ownership (CSMAR; -0.53pp, not significant at the 10% level), and QFII-specific foreign ownership (-0.017pp, not significant at the 10% level). The QFII data strongly support the structural-exposure interpretation, since sanctioned A-share firms held only 0.03% QFII pre-shock, leaving essentially no foreign capital to lose. Two qualifications temper this categorical null. A dose-response test among the most-exposed firms yields a significant negative effect (-10.73, significant at the 5% level), which suggests a weak, transitory channel in the subset with the highest pre-shock U.S.-nexus exposure, while the top-10 analysis shows a small increase in Hong Kong nominee/HKSCC clearing-account ownership ($+0.16$, significant at the 5% level), likely reflecting Stock Connect reallocation. These signals do not survive as robust aggregate findings. The mainland null is consistent with a capital-base substitution channel that operates only where baseline foreign exposure is substantial, which is why a measurable effect appears only among explicitly sanctioned Hong Kong-listed firms, where baseline international ownership stood at approximately 26%.

The Hong Kong surname classifier is a noisy proxy, in that it cannot distinguish a Hong Kong permanent resident with a Western surname from a U.S. citizen, and it misclassifies ethnically Chinese directors who hold Western citizenship. Our multi-dialect approach reduces this error without eliminating it, and the independent CSMAR-sourced nationality classifier (70.5% coverage of Hong Kong director-years, with a 25-firm overlap with the surname group) supplies the triangulating cross-check that we rely on throughout. Because the explicit-sanctions sample comprises only 11 firms, all of them large SOEs, the strong ownership results may not generalize to smaller or privately owned firms, and the Stock Connect sign reversal between sanctioned and intent-to-treat firms further complicates the ownership interpretation. Several of the cross-market statements we retain lean on the eleven-firm coefficient, and we accordingly report them tentatively.

Several extensions would strengthen the analysis. With the within-board exit-hazard design above now differencing out firm-level mean reversion directly, an explicit director-tenure control is the main remaining refinement to the identification battery. Firm-level QFII/RQFII registry data, in place of the top-10 approximations, would sharpen the capital-base test. The director changes file contains exact appointment and departure dates that could support event-time designs centered on each firm's own NS-CMIC or Entity List designation date; if the within-board exit gap opened around each firm's individual designation and not uniformly in 2021, the effect would be attributed more tightly to the sanctions events themselves. A Callaway–Sant'Anna (2021) estimator would again be the appropriate tool should those designation dates prove to constitute genuinely staggered treatment timing.

References

Adams, R. B., Hermalin, B. E., and Weisbach, M. S. (2010). The role of boards of directors in corporate governance: A conceptual framework and survey. *Journal of Economic Literature*, 48(1):58–107.

Berkowitz, D., Lin, C., and Liu, S. (2022). De-politicization and corporate transformation: Evidence from china. *Journal of Law, Economics, and Organization*, 38(2):479–510.

Biglaiser, G. and Lektzian, D. (2011). The effect of sanctions on U.S. foreign direct investment. *International Organization*, 65(3):531–551.

Cao, F., Sun, J., and Yuan, R. (2019). Board directors with foreign experience and stock price crash risk: Evidence from china. *Journal of Business Finance & Accounting*, 46(9–10):1144–1170.

Chen, J., Garel, A., and Tourani-Rad, A. (2019). The value of academics: Evidence from academic independent director resignations in china. *Journal of Corporate Finance*, 58:393–414.

Doidge, C., Karolyi, G. A., and Stulz, R. M. (2004). Why are foreign firms listed in the U.S. worth more? *Journal of Financial Economics*, 71(2):205–238.

Egger, P. H. and Zhu, J. (2020). The US–Chinese trade war: An event study of stock-market responses. *Economic Policy*, 35(103):519–559.

Fahlenbrach, R., Low, A., and Stulz, R. M. (2010). The dark side of outside directors: Do they quit when they are most needed? *Journal of Financial Economics*, 98(3):400–414.

Fan, J. (2021). The effect of regulating political connections: Evidence from china's board of directors ban. *Journal of Comparative Economics*, 49(2):553–578.

Firth, M., Wong, S. M. L., Xin, Q., and Yick, H. Y. (2016). Regulatory sanctions on independent directors and their consequences to the director labor market: Evidence from china. *Journal of Business Ethics*, 134(4):693–708.

Giannetti, M., Liao, G., and Yu, X. (2015). The brain gain of corporate boards: Evidence from china. *The Journal of Finance*, 70(4):1629–1682.

Gong, M., Wang, Y., and Yang, X. (2021). Do independent directors restrain controlling shareholders' tunneling? evidence from a natural experiment in china. *Economic Modelling*, 94:548–559.

He, C., Chen, R., and Liu, Y. (2022). US–China trade war and China's stock market: An event-driven analysis. *Economic Research–Ekonomska Istraživanja*, 35(1):3277–3290.

He, W. and Liu, W. (2016). Why do companies appoint independent directors having professional legal expertise? *China Journal of Accounting Studies*, 4(4):379–405.

Hu, H., Yang, S., Zeng, L., and Zhang, X. (2024). U.S.–China trade conflicts and R&D investment: Evidence from the BIS entity lists. *Humanities and Social Sciences Communications*, 11(1):829.

Jiang, G., Lee, C. M. C., and Yue, H. (2010). Tunneling through intercorporate loans: The china experience. *Journal of Financial Economics*, 98(1):1–20.

Jiang, W., Wan, H., and Zhao, S. (2016). Reputation concerns of independent directors: Evidence from individual director voting. *The Review of Financial Studies*, 29(3):655–696.

Liu, B. and Li, Q. (2023). US export control to China and China enterprise innovation. *Journal of Finance and Economics*, 49(12):19–33.

Liu, X., Zhou, A. J., Zhou, S. S., Bai, T., and Tan, X. (2025). Shock anticipation: Voluntary turnover by foreign directors in response to sanctions on peer firms. *Journal of Management Studies*.

Liu, Y., Miletkov, M. K., Wei, Z., and Yang, T. (2015). Board independence and firm performance in china. *Journal of Corporate Finance*, 30:223–244.

Masulis, R. W., Wang, C., and Xie, F. (2012). Globalizing the boardroom—the effects of foreign directors on corporate governance and firm performance. *Journal of Accounting and Economics*, 53(3):527–554.

Miletkov, M. K., Poulsen, A. B., and Wintoki, M. B. (2017). Foreign independent directors and the quality of legal institutions. *Journal of International Business Studies*, 48(2):267–292.

Oh, M. and Kim, D. (2024). Effect of the U.S.–China trade war on stock markets: A financial contagion perspective. *Journal of Financial Econometrics*, 22(4):954–1005.

Oxelheim, L., Gregorič, A., Randøy, T., and Thomsen, S. (2013). On the internationalization of corporate boards: The case of nordic firms. *Journal of International Business Studies*, 44(3):173–194.

Oxelheim, L. and Randøy, T. (2003). The impact of foreign board membership on firm value. *Journal of Banking & Finance*, 27(12):2369–2392.

Tao, Q., Wei, K. C. J., Xiang, E., and Yi, B. (2022). Board directors' foreign experience and firm dividend payouts. *Journal of Corporate Finance*, 75:102218.

Tian, X., Xu, Y., and Yiu, D. W. (2025). Foreign director exit in the midst of deteriorating bilateral political relations. *Journal of Management*, 51(8):3426–3456.

Yu, L., Bai, T., Yin, J., and Tan, X. (2024). Overcoming the liability of origin by hiring foreign independent directors: Evidence from chinese firms' cross-border m&as. *Journal of World Business*, 59(2):101520.

Zhu, J., Ye, K., Tucker, J. W., and Chan, K. C. (2016). Board hierarchy, independent directors, and firm value: Evidence from china. *Journal of Corporate Finance*, 41:262–279.

A Conceptual Framework

Figure A1 consolidates this Article's identification logic and the compliance-chill mechanism in a single diagram. It traces the quasi-exogenous sanctions shock, the individual-liability channel through which it operates, the director exit we interpret as causal, the descriptive board recomposition that follows, and the selective-versus-indiscriminate threshold around which the cross-market evidence is organized.

B Robustness and Pre-Trend Diagnostics

B.1 Treatment-Definition Robustness

Table A1: Treatment-Definition Robustness: Main Results and Placebo Tests

	v5 (CSMAR) N_T=1,109	v7 (+scraper nat.) N_T=1,164	v8 (+scraper 3-prong) N_T=1,298
Panel A: Main Results			
Director exit (share)	-0.0200^{***}	-0.0190^{***}	-0.0180^{***}
	(<0.001)	(<0.001)	(<0.001)
Director exit (count)	-0.3692^{***}	-0.3487^{***}	-0.3375^{***}
	(<0.001)	(<0.001)	(<0.001)
Replacement (domestic technocrat)	$+0.0108^{**}$	$+0.0103^{**}$	$+0.0100^{**}$
	(0.024)	(0.024)	(0.027)
Panel B: Placebo Tests			
Placebo T1 (same firms)	+0.0007	+0.0010	+0.0011
	(0.484)	(0.319)	(0.279)
Placebo T2 (unexposed)	-0.0161^{***}	-0.0165^{***}	-0.0169^{***}
	(<0.001)	(<0.001)	(<0.001)

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. v5 uses the CSMAR three-prong heuristic; v7 adds Baidu-scraper-classified U.S. nationals; v8 adds scraper work experience and licenses. Placebo T1 uses non-U.S. foreign share at treated firms, and T2 uses non-U.S.-foreign-treated firms with no U.S. exposure. All specifications include firm and year fixed effects, controls, and firm-clustered standard errors.

Table A1 reports our results under three treatment definitions. The primary specification (v5) relies on CSMAR structured data to classify U.S.-nexus directors via the three-prong heuristic, identifying 1,109 treated firms. The expanded definitions add directors recovered from Baidu search snippets, so that v7 incorporates scraper-classified U.S. nationals (+55 firms), while v8 further adds directors with U.S. work experience or U.S. licenses (+134 firms).

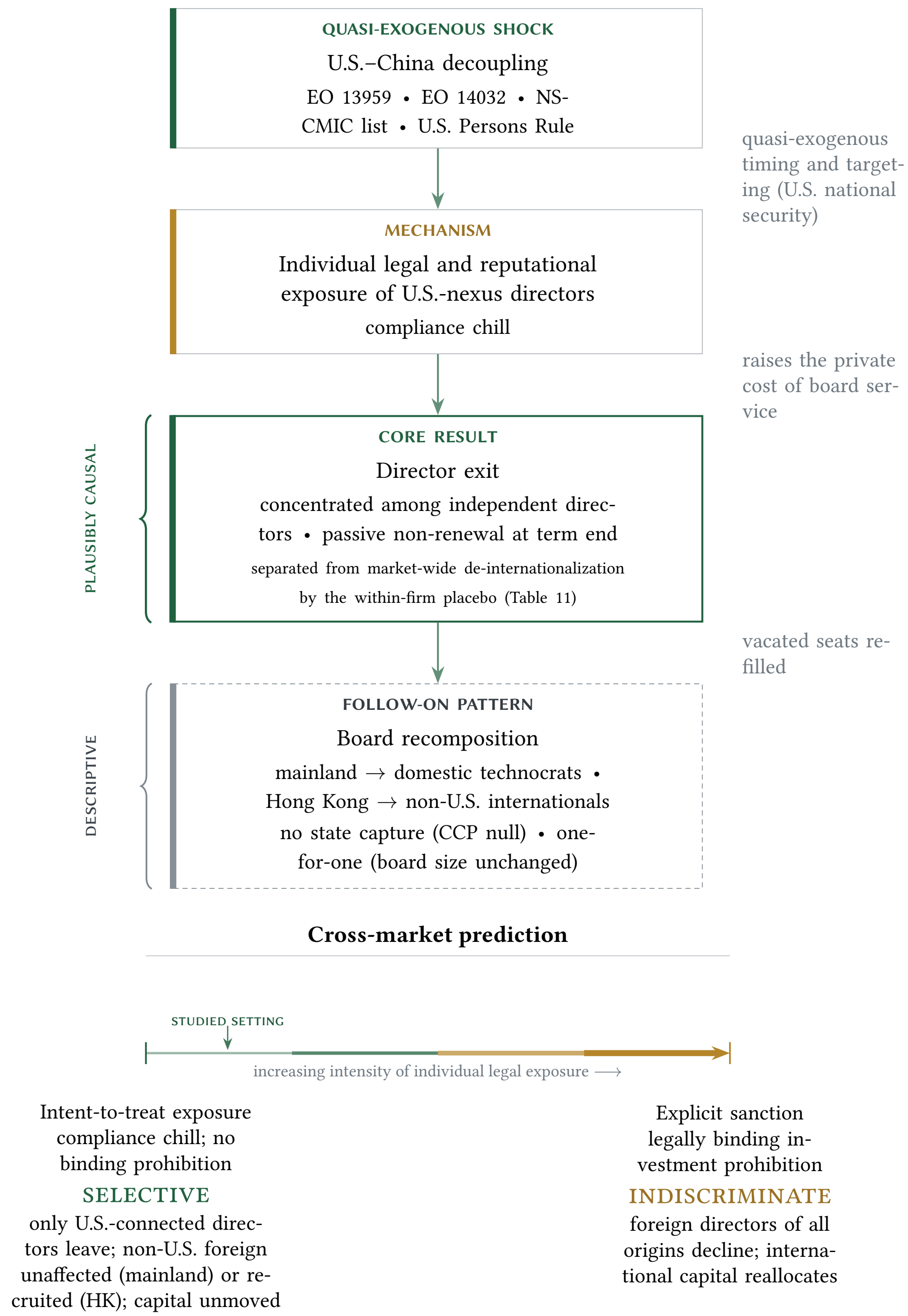

Figure A1: Identification and mechanism. The vertical chain traces the causal architecture: a quasi-quasi-exogenous U.S. sanctions shock raises the individual legal and reputational exposure of U.S.-nexus directors (compliance chill), which we interpret as causing the director exit, around which boards descriptively recompose. The within-firm placebo decomposition (Table 11)—under which only U.S.-connected directors leave while non-U.S. foreign directors at the same firms do not—separates the U.S.-specific channel from a concurrent market-wide de-internationalization and is the feature of the design least vulnerable to firm-level mean reversion. The lower panel states the central cross-market prediction: the response is selective under intent-to-treat exposure and becomes indiscriminate once an explicit sanction renders the holding legally prohibited.

We adopt v5 as our primary specification because the v7/v8 additions introduce measurement noise. The scraper-identified directors differ systematically from the CSMAR-identified population: they are predominantly non-independent (75–78% vs. 52%), they exhibit weaker post-shock exit rates (-25.5% vs. -41.9% at treated firms over 2020–2024), and 73–74% have zero observations in the analysis window, having served before the sanctions period. These characteristics are consistent with the scraper capturing less prominent directors whose U.S. connections are harder to verify from structured data.

Our results remain robust across all three definitions. Panel A shows that the exit estimate attenuates monotonically from -0.020^{***} (v5) to -0.018^{***} (v8), consistent with adding firms whose U.S.-nexus directors exit at lower rates. The replacement result is stable ($+0.011^{**}$ to $+0.010^{**}$), while Panel B confirms that the placebo tests are invariant to the treatment definition.

B.2 Cross-Specification Robustness Summary

Table A2: Cross-Specification Robustness Summary

	DV	$\hat{\beta}$	p	N	Sample
Panel A: Main Results					
H1	U.S.-nexus director share	-0.0200^{***}	<0.001	17,171	2019–2024
H2	Domestic technocrat	$+0.0108^{**}$	0.024	314,050	2019–2024
Panel B: Extensions					
Ext 1a (dose)	U.S.-nexus director share	-0.3692^{***}	<0.001	17,171	2019–2024
Ext 1b	Euphemistic resignation	-0.0652^{**}	0.030	54,117	2019–2024
Ext 1c (Independent)	U.S.-nexus share (independent)	-0.0477^{***}	<0.001	17,171	2019–2024
Ext 2a (Technical-background index)	Technical-background index	$+0.0058^{**}$	0.011	314,050	2019–2024
Ext 2b (Board size)	Board size	$+0.0090$	0.821	17,110	2019–2024
Ext 2b (Independent ratio)	Independent-director ratio	-0.0859	0.611	17,109	2019–2024
Ext 2c (POE indep)	Log(indep. director pay)	$+0.1596^{**}$	0.028	5,437	2019–2024
Ext 2c (TOP3, SOE)	Log(top-3 exec. pay)	-0.0574^{**}	0.022	5,467	2019–2024
Panel C: Industry $\times$ Year FE Robustness					
H1 (Ind×Yr)	U.S.-nexus director share	-0.0198^{***}	<0.001	17,171	2019–2024
Panel D: Alternative Clustering (Most Conservative)					
H1 (Firm+Year)	U.S.-nexus director share	-0.0200^{***}	<0.001	17,171	2019–2024
H2 (Firm)	Domestic technocrat	$+0.0108^{**}$	0.024	314,050	2019–2024

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. All specifications include firm and year fixed effects unless noted; the replacement analysis uses replacement-pair data with the treated-U.S.-exit indicator as the treatment variable.

Table A2 consolidates the key coefficient estimates across our retained hypotheses and extensions. Panel A reports the main results, whereas Panel B reports the extensions, including the corrected board-size result ($+0.009$, null) and the independent-director ratio (-0.086, null), together with the SOE executive-pay decline (-0.057, significant at the 5% level). Panel C reports industry $\times$ year fixed-effect specifications, under which the exit result is virtually unchanged (-0.020***). Panel D reports the most conservative clustering: the exit result remains highly stable under two-way firm and year clustering, where the standard error increases fourfold yet remains significant at the 1% level, and the replacement result survives. The wild cluster bootstrap likewise confirms the exit result (significant at the 1% level) under both industry (18 clusters) and province (36 clusters) bootstraps.

B.3 International Ownership: Intent-to-Treat

Table A3: International Ownership: Intent-to-Treat (Hong Kong Quarterly Panel)

	(1)	(2)	(3)	(4)	(5)	(6)
	DV: International ownership (%)		DV: Mainland ownership (%)		DV: Stock Connect ownership (%)	
	Western	Foreign	Western	Foreign	Western	Foreign
Treated × Post	+2.42	+0.97	+1.36	−0.51	-2.47^{***}	-3.12^{***}
	(0.226)	(0.215)	(0.227)	(0.558)	(<0.001)	(<0.001)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Firm	Firm	Firm	Firm	Firm	Firm
Observations	71,510	71,510	71,509	71,509	20,120	20,120
Firms	2,307	2,307	2,307	2,307	739	739
R^2 (within)	0.0078	0.0113	0.0005	0.0004	0.0601	0.0170

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The table reports broader intent-to-treat ownership estimates for Western-surname and foreign-dialect treatment definitions in the Hong Kong quarterly panel.

Table A3 extends the ownership analysis to the broader intent-to-treat sample, using the Western-surname and foreign-dialect treatment definitions. The results are uniformly null for international and Chinese ownership, since none of the four coefficients approaches significance (not significant at the 10% level). The only significant result is a negative Stock Connect effect (-2.47, significant at the 1% level for Western treatment; -3.12, significant at the 1% level for foreign-dialect). This sign reversal relative to the explicitly sanctioned firms (Table 9) likely reflects compositional differences. Explicitly sanctioned firms are predominantly large SOEs that attract Stock Connect inflows as mainland investors substitute for departing international holders, whereas the broader intent-to-treat sample includes smaller firms in which foreign-director presence signals international orientation without triggering forced ownership reallocation. Ownership concentration (top-10 shareholder share) is null across all specifications (not significant at the 10% level). These nulls constitute the financial-capital half of the threshold result: measurable capital reallocation appears to require the binding prohibition of explicit sanctions, while mere geopolitical exposure yields null ownership effects and limited evidence because baseline foreign exposure is low.

B.4 Pre-Trend Diagnostics

Our preferred mainland specification (2019–2024, Post $\geq$ 2021) has a deliberately thin pre-period. The single available pre-treatment coefficient (2020 relative to the 2019 base year) is $+0.0003$ (not significant at the 10% level), showing no detectable pre-shock divergence; parallelism cannot be assessed from one pre-period and is not the design's identifying assumption, with the within-firm placebo (Table 11) supplying the operative falsification against mean reversion. The event study shows a monotonic post-shock decline from -0.009 (2021) to -0.032^{***} (2024). The extended 2018–2024 mainland event study reveals a significant 2018 coefficient (-0.0075, significant at the 1% level), which we trace to 140 "early-treated" firms with U.S.-nexus departures coincident with the Section 301 trade-war tariffs (July 2018); excluding these firms eliminates the anomaly. In Hong Kong, the surname-based event study shows significant pre-trend coefficients in 2015 (-0.030^{***}) and 2016 (-0.020^{**}), yet the relevant 2017–2019 pre-treatment window is clean (all not significant at the 10% level); the earlier anomaly likely reflects the onset of board internationalization among Hong Kong-listed firms in the mid-2010s. The nationality-based event study displays a similar pattern, with 2017–2019 clean (all not significant at the 10% level) and a monotonic post-shock decline reaching -0.029^{***} by 2024.

B.5 Within-Board Exit Hazard

Table A4 reports the director-level exit-hazard estimates that underlie Figure 3. The dependent variable indicates that a director leaves the board between year t and $t+1$, defined on director-years whose firm is still listed in $t+1$ so that delistings are not counted as exits. Column 1 is a pooled linear-probability model; column 2 adds firm and year fixed effects; columns 3–5 absorb firm $\times$ year fixed effects, which restrict identification to the contrast between U.S.-connected directors and their non-U.S. colleagues on the same board in the same year and thereby difference out any firm-level mean reversion. The coefficient on U.S.-nexus $\times$ Post is positive in every specification and attains significance once the growing dynamic effect is not averaged against a near-zero 2020 base, reaching $+0.030^{***}$ over the full 2018–2023 window (column 4) and $+0.038^{***}$ within treated firms (column 5). The positive U.S.-nexus main effect measures the pre-period within-board exit gap, which the event study in Figure 3 shows to have no positive pre-trend before 2021.

Table A4: Director-Level Exit Hazard with Firm $\times$ Year Fixed Effects

	(1) Pooled	(2) Firm + Year FE	(3) Firm $\times$ Year FE	(4) Firm $\times$ Year (2018–23)	(5) Firm $\times$ Year (treated)
U.S.-nexus $\times$ Post	+0.0122	+0.0118	+0.0146	+0.0295***	+0.0379***
	(0.228)	(0.243)	(0.146)	(0.001)	(<0.001)
U.S.-nexus	+0.0178**	+0.0213***	+0.0186***	+0.0037	+0.0186***
	(0.014)	(0.003)	(0.010)	(0.513)	(0.010)
Firm FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Firm $\times$ Year FE	No	No	Yes	Yes	Yes
Window	2019–23	2019–23	2019–23	2018–23	2019–23
Sample	All	All	All	All	Treated
Mean exit rate	0.145	0.145	0.145	0.144	0.149
Director-years	272,817	272,817	272,817	323,098	105,309
Firms (clusters)	2,934	2,934	2,934	2,941	1,107

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Values in parentheses are p-values. The dependent variable is an indicator that a director leaves the board between year t and $t+1$; director-years whose firm is not listed in $t+1$ are dropped so that delistings are not counted as exits. Post $= 1$ for $t \geq 2021$. Columns 3–5 absorb firm $\times$ year fixed effects, so identification rests on the within-board contrast between U.S.-nexus and non-U.S. directors; column 4 uses the full 2018–2023 window and column 5 restricts the sample to treated firms. Standard errors are clustered by firm.

Acknowledgments

The work described in this paper was fully supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China [Project No.: HKU 14601025].